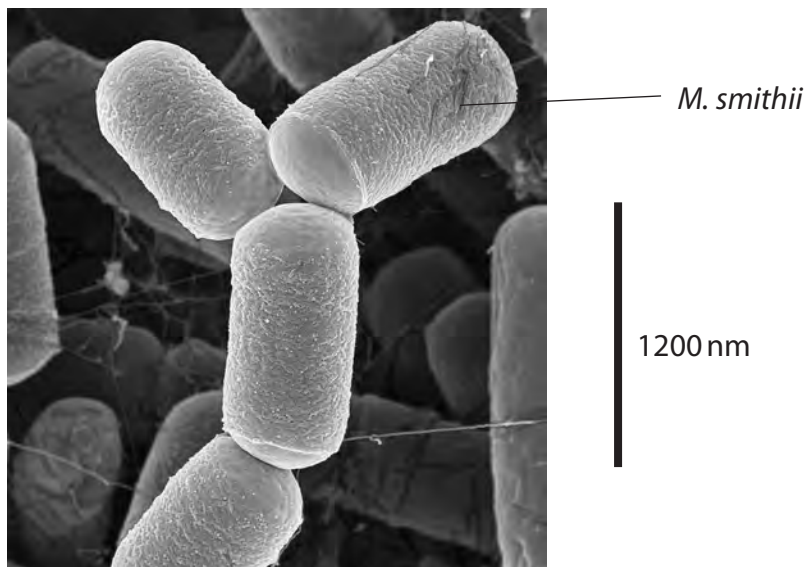


- 2 *Methanobrevibacter smithii* (*M. smithii*) is a single-celled microorganism found in the human intestine.

The photograph shows *M. smithii*.



(© DENNIS KUNKEL MICROSCOPY / SCIENCE PHOTO LIBRARY)

- (a) (i) Calculate the magnification of this photograph.

Give your answer in standard form to two significant figures.

(2)

Answer

- (ii) Explain which type of microscope was used to take this photograph.

(2)

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(b) Scientists classified this microorganism as a species in the domain Archaea.

- (i) Describe the information the scientists would have used to classify *M. smithii* into the Archaea domain.

(2)

- (ii) Scientists have identified a similar microorganism in the human mouth.

This microorganism is called *Methanobrevibacter oralis* (*M. oralis*).

Explain how the scientists could confirm that *M. smithii* and *M. oralis* are different species of Archaea.

(3)

(Total for Question 2 = 9 marks)



P 6 5 8 1 1 A 0 5 2 8